# Discrete differential geometry in homotopy type theory

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October 2025

# Summary

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We make some constructions in HoTT that deserve to be called geometry:

- connections, curvature, and vector fields
- the index of a vector field
- a theorem in dimension 2 that total curvature $=$ total index

# Classical $\to$ HoTT

Let $M$ be a compact, smooth, oriented 2-manifold without boundary, $F_A$ the curvature of a connection $A$ on the tangent bundle, and $X$ a vector field with isolated zeroes $x_1, \ldots, x_n$.

$$\frac{1}{2\pi}\int_M F_A = \sum_{i=1}^{n} \text{index}_X(x_i) = \chi(M) \qquad \leftarrow \text{Gauss-Bonnet/Poincaré-Hopf}$$

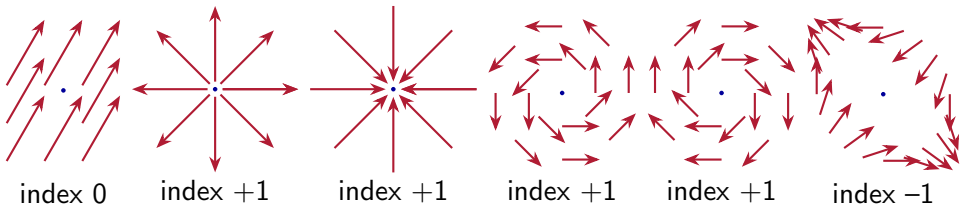
$$\downarrow \qquad\qquad \downarrow$$

$$\sum_{\text{faces } F} \flat_F = = = \sum_{\text{faces } F} L_F^X \qquad \leftarrow \text{our theorem}$$

# Classical index

Near an isolated zero there are only three possibilities: index 0, 1, –1.

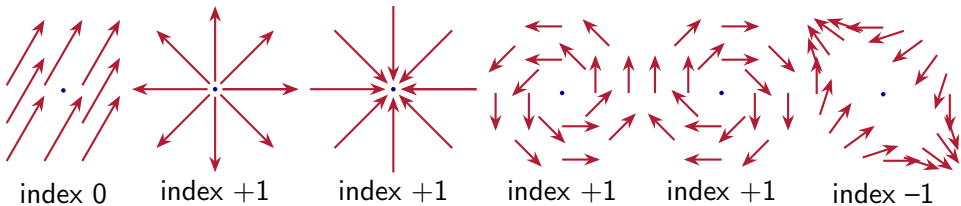
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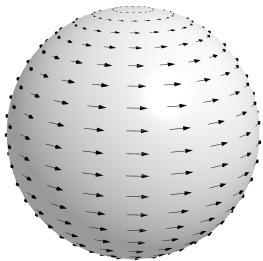



(This definition is using an implicit ad-hoc connection near the zero, more on this later.)

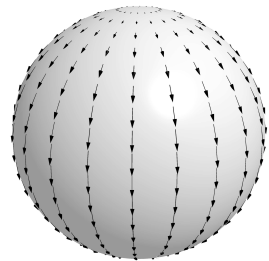
# Poincaré-Hopf theorem

The total index of a vector field is the Euler characteristic.

Examples:



Rotation: index $+1$ at each pole $=$ **2**



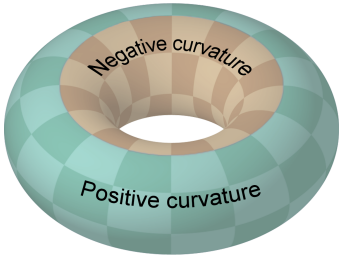
Height: index $+1$ at each pole $=$ **2**

# Gauss-Bonnet theorem

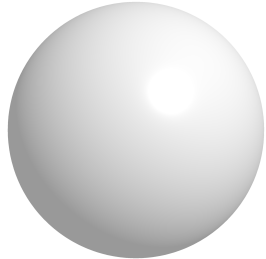
Total curvature divided by $2\pi$ is the Euler characteristic.

Curvature in 2D is a function $F_A : M \to \mathbb{R}$.

$\int_M F_A$ sums the values at every point.




Positive and negative curvature cancel: **0**



Constant curvature 1, area $4\pi$: **2**

# Plan

- Combinatorial manifolds
- Torsors and classifying maps
- Connections and curvature
- Vector fields
- Main theorem

# HoTT background


1. Symmetry,
   Bezem, M., Buchholtz, U., Cagne, P., Dundas, B. I., and Grayson, D. R., (2021-)
   `https://github.com/UniMath/SymmetryBook`.
2. Central H-spaces and banded types,
   Buchholtz, U., Christensen, J. D. , Flaten, J. G. T., and Rijke, E. (2023)
   arXiv:2301.02636
3. Nilpotent types and fracture squares in homotopy type theory,
   Scoccola, L. (2020)
   MSCS 30(5). arXiv:1903.03245

# Combinatorial manifolds

# Manifolds in HoTT

- Recall the classical theory of simplicial complexes
- Define a realization procedure to construct types

# Simplicial complexes

**Definition**

An abstract simplicial complex $M$ of dimension $n$ is an ordered list of sets $M \stackrel{\text{def}}{=} [M_0, \ldots, M_n]$ consisting of

- a set $M_0$ of vertices
- sets $M_k$ of $k$-faces: subsets of $M_0$ of cardinality $k+1$
- downward closed: if $F \in M_k$ and $G \subseteq F$, $|G| = j+1$ then $G \in M_j$

We call the truncated list $M_{\leq k} \stackrel{\text{def}}{=} [M_0, \ldots, M_k]$ the $k$-skeleton of $M$.

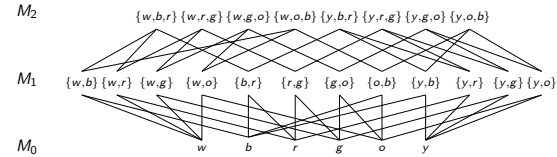
# Simplicial complexes

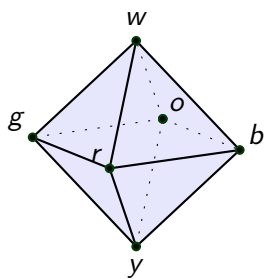
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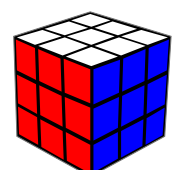
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# Simplicial complexes

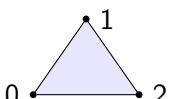
## Example

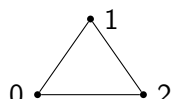
The complete simplex of dimension $n$, denoted $\Delta(n)$, is the set $\{0, \ldots, n\}$ and its power set. The $(n-1)$-skeleton $\Delta(n)_{\leq(n-1)}$ is denoted $\partial\Delta(n)$ and will serve as a combinatorial $(n-1)$-sphere.

$\Delta(1)$ is visually 

, $\partial\Delta(1)$ is visually 

,

$\Delta(2)$ is visually 

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# Homotopy realization: dimension 0

We will realize simplicial complexes by means of a sequence of pushouts.

It is likely we are exactly applying the $\int$ modality, but we won't need that.

Base case: the realization $\mathbb{M}$ of a 0-dimensional complex $M$ is $M_0$.

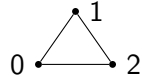
In particular the 0-sphere $\partial\Delta(1) \overset{\text{def}}{=} \partial\Delta(1)_0$.

# Homotopy realization: dimension 1

For a 1-dim complex $M \overset{\text{def}}{=} [M_0, M_1]$ the realization is given by

$$\begin{array}{ccc} M_1 \times \partial\Delta(1) & \xrightarrow{\mathsf{pr}_1} & M_1 \\ {\scriptstyle \mathbb{A}_0}\downarrow & \overset{h_1}{\nearrow} & \downarrow{\scriptstyle *_{\mathbb{M}_1}} \\ M_0 = \mathbb{M}_0 & \longrightarrow & \mathbb{M}_1 \end{array}$$

# Homotopy realization: dimension 1

For example the simplicial 1-sphere $\partial\mathbb{\Delta}(2) \overset{\text{def}}{=}$ 

is given by

$$\begin{array}{ccc} \partial\Delta(2)_1 \times \partial\mathbb{\Delta}(1) & \longrightarrow & \partial\Delta(2)_1 \\ \downarrow & \overset{h_1}{\nearrow} & \downarrow \\ \partial\Delta(2)_0 & \longrightarrow & \partial\mathbb{\Delta}(2) \end{array}$$

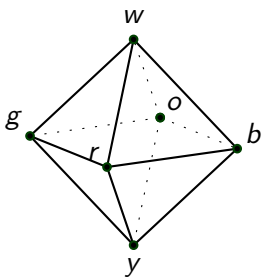
i.e.

$$\begin{array}{ccc} \{\{0,1\},\{1,2\},\{2,0\}\} \times \{0,1\} & \longrightarrow & \{\{0,1\},\{1,2\},\{2,0\}\} \\ \downarrow & \overset{h_1}{\nearrow} & \downarrow \\ \{0,1,2\} & \longrightarrow & \partial\mathbb{\Delta}(2) \end{array}$$

# Homotopy realization: dimension 1

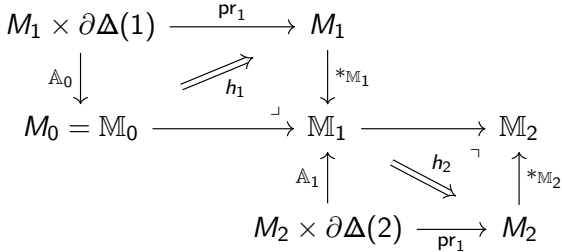
Or the 1-skeleton of the octahedron $\mathbb{O}$:

$$\begin{array}{ccc} \{\{w,g\},\ldots\} \times \{0,1\} & \longrightarrow & \{\{w,g\},\ldots\} \\ \downarrow & \overset{h_1}{\nearrow} & \downarrow \\ \{w,g,\ldots\} & \longrightarrow & \mathbb{O}_1 \end{array}$$



# Homotopy realization: dimension 2

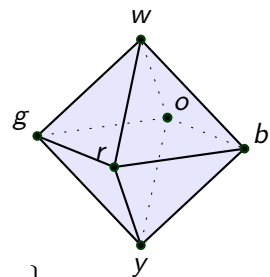
To realize $M \overset{\text{def}}{=} [M_0, M_1, M_2]$ use $\partial\Delta(1), \partial\Delta(2)$:



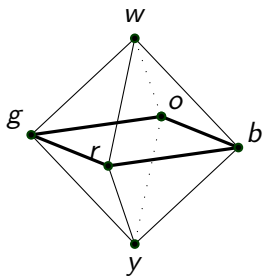
# Homotopy realization: dimension 2

The full octahedron $\mathbb{O}$:

$$\begin{array}{ccccc}
\{\{w,g\},\ldots\}\times\{0,1\} & \xrightarrow{\mathrm{pr}_1} & \{\{w,g\},\ldots\} & & \\
\downarrow & \overset{h_1}{\Rightarrow} & \downarrow & & \\
\{w,g,\ldots\} & \longrightarrow & \mathbb{O}_1 & \longrightarrow & \mathbb{O}_2 \\
 & & \uparrow & \overset{h_2}{\Rightarrow} & \uparrow \\
 & & \{\{w,g,r\},\ldots\}\times\partial\Delta(2) & \xrightarrow[\mathrm{pr}_1]{} & \{\{w,g,r\},\ldots\}
\end{array}$$



# Homotopy realization: dimension 2




$\leftarrow \text{link}(w)$

The link of a vertex $w$ in a 2-complex is: the faces not containing $w$ but whose union with $w$ is a face.

A combinatorial $n$-manifold is a simplicial complex all of whose links are* simplicial $(n-1)$-spheres.

Links will be our model of the tangent space.

*the (classical) geometric realization is homeomorphic to a sphere

# Combinatorial manifolds $\leftrightarrow$ smooth manifolds

## Theorem (Whitehead (1940))

*Every smooth n-manifold has a compatible structure of a combinatorial manifold: a simplicial complex of dimension n such that the link is a combinatorial $(n-1)$-sphere, i.e. its geometric realization is an $(n-1)$-sphere.*

`https://ncatlab.org/nlab/show/triangulation+theorem`

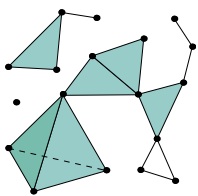
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Counterexample: Wikipedia says this is a simplicial complex, but we can see it fails the link condition:



# Torsors

What type families $\mathbb{M} \to \mathcal{U}$ will we consider? Families of torsors, also called principal bundles.

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Let $G$ be a (higher) group.

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- Let $BG$ be the type of $G$-torsors.
- Let $G_{\mathrm{reg}}$ be the $G$-torsor consisting of $G$ acting on itself on the right.

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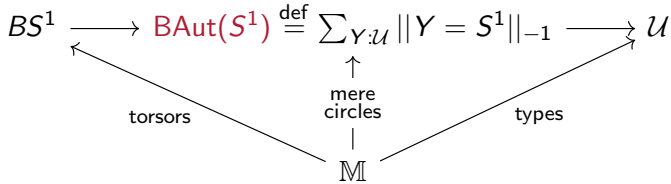
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See the Buchholtz et al. H-spaces paper for more.

# How to map into $BS^1$

To construct maps into $BS^1$ we lift a family of mere circles.

families of:

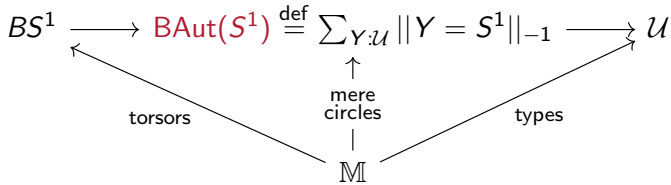



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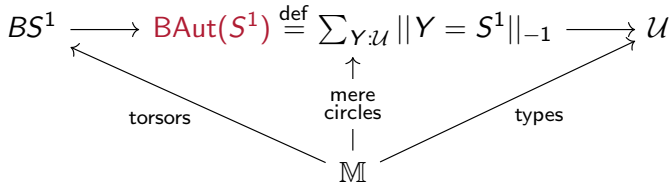

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Other names:

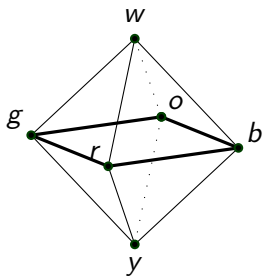
- $\mathrm{BAut}(S^1) = BO(2) = \mathrm{EM}(\mathbb{Z}, 1)$ (where $\mathrm{EM}(G, n) \stackrel{\mathrm{def}}{=} \mathrm{BAut}(\mathrm{K}(G, n))$)
- $BS^1 = BSO(2) = \mathrm{K}(\mathbb{Z}, 2)$

# Connections and curvature

# Connections

Connections are extensions of a bundle to higher skeleta.

# Recall link



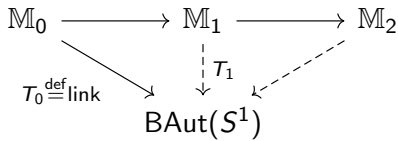

$\leftarrow \mathrm{link}(w)$

The link of a vertex $w$ in a 2-complex is: the faces not containing $w$ but whose union with $w$ is a face.

Define the tangent bundle on a combinatorial manifold to be $T_0 \stackrel{\text{def}}{=} \mathrm{link} : \mathbb{M}_0 \to \mathrm{BAut}(S^1)$.

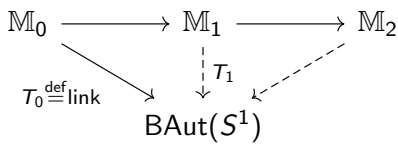
# Connections on the tangent bundle

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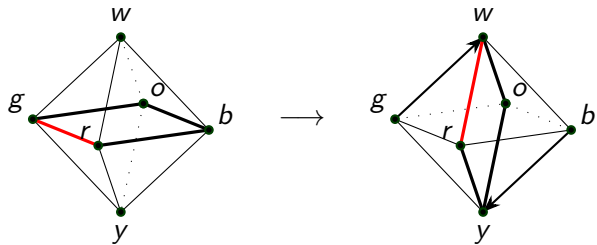

The classical definition is a function $(\gamma : [0,1] \to M) \mapsto (\text{tr}_\gamma : P(\gamma(0)) \simeq P(\gamma(1)))$.

HoTT provides the motivation for asking: what does the bundle do on paths?

# $T_1 : \mathbb{M}_1 \to \mathrm{BAut}(S^1)$ extending link

We will define $T_1$ on the edge $wb$, so we need a term $T_1(wb) : \mathrm{link}(w) =_{\mathrm{BAut}(S^1)} \mathrm{link}(b)$.

We imagine tipping:

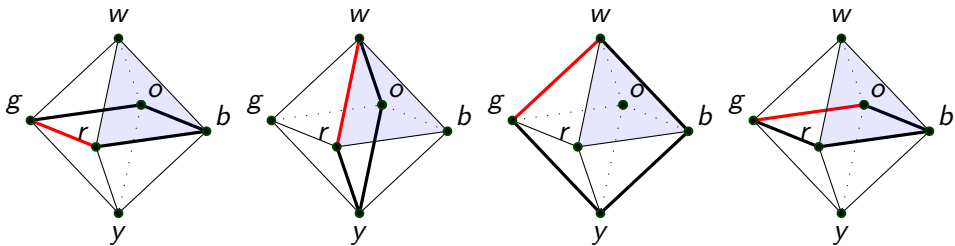



$T_1(g : \mathrm{link}(w)) \stackrel{\mathrm{def}}{=} w : \mathrm{link}(b), \ldots$.

Use this method to define $T_1$ on every edge.

# $T_1 : \mathbb{M}_1 \to \mathrm{BAut}(S^1)$ extending link

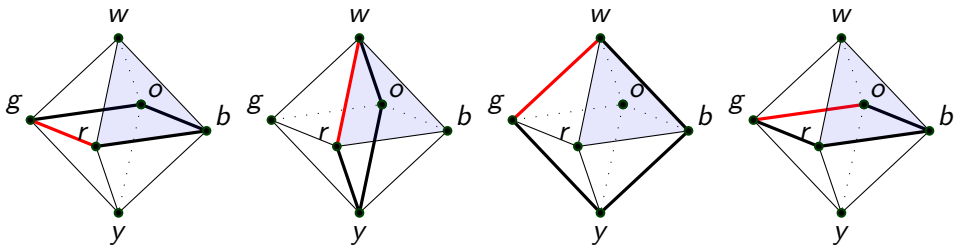
Denote the path $wb \cdot br \cdot rw$ by $\partial(wbr)$. Consider $T_1(\partial(wbr))$:




We come back rotated by 1/4 turn. Call this rotation $R : \mathrm{link}(w) =_{BAut(S^1)} \mathrm{link}(w)$.

# Extending $T_1$ to a face

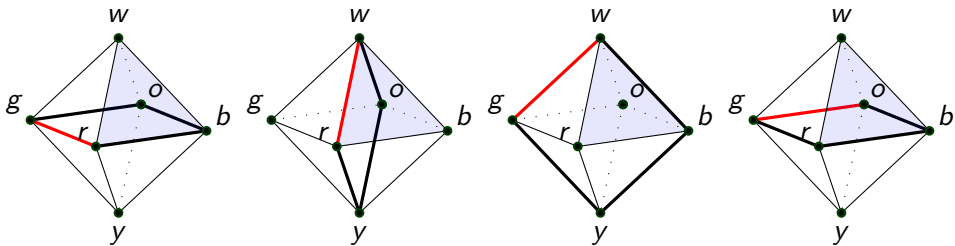
Let $H_{wbr} : \mathrm{refl}_w =_{w =_{\mathbb{M}} w} \partial(wbr)$ be the filler homotopy of the face.



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$T_2$ must live in $T_1(\mathsf{refl}_w) =_{(\mathsf{link}(w) =_{\mathsf{BAut}(S^1)} \mathsf{link}(w))} T_1(\partial(wbr)) = R$

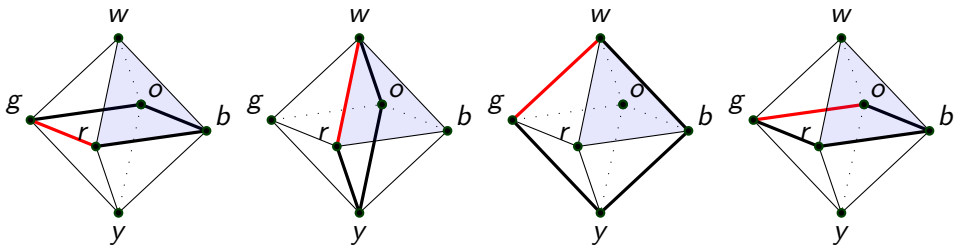


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Let $H_{wbr} : \mathsf{refl}_w =_{w =_{\mathbb{M}} w} \partial(wbr)$ be the filler homotopy of the face.

$T_2$ must live in $T_1(\mathsf{refl}_w) =_{(\mathsf{link}(w) =_{\mathsf{BAut}(S^1)} \mathsf{link}(w))} T_1(\partial(wbr)) = R$

$T_2$ must be a homotopy $H_R : \mathsf{id} = R$ between automorphisms of $\mathsf{link}(w)$.



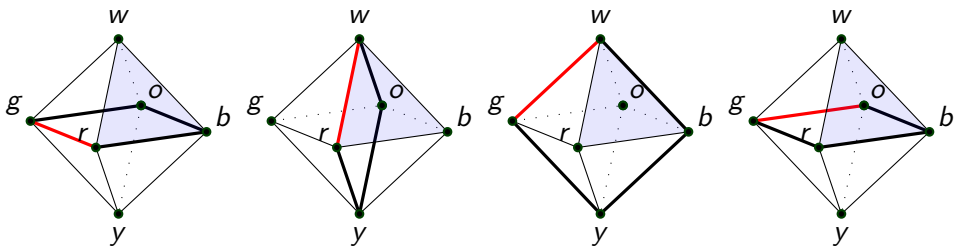
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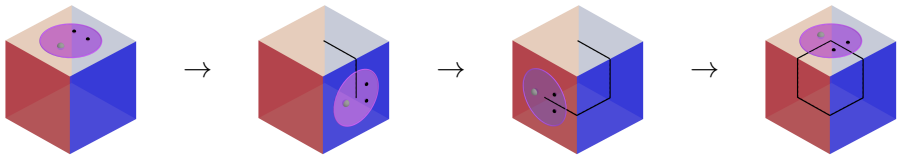
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For example, a path $H_R(g) : g = Rg = o$. Choose $go$.



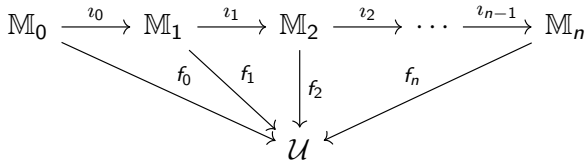
# Original inspiration



# The definition of a connection

## Definition

If $\mathbb{M} \stackrel{\text{def}}{=} \mathbb{M}_0 \xrightarrow{\imath_0} \cdots \xrightarrow{\imath_{n-1}} \mathbb{M}_n$ is the realization of a combinatorial manifold and all the triangles commute in the diagram:

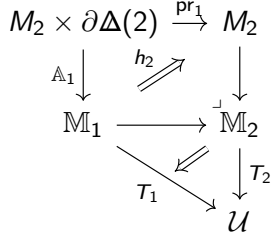



- The map $f_k$ is a $k$-bundle on $\mathbb{M}$.
- The pair given by the map $f_k$ and the proof $f_k \circ \imath_{k-1} = f_{k-1}$, i.e. that $f_k$ extends $f_{k-1}$ is called a $k$-connection on the $(k-1)$-bundle $f_{k-1}$.

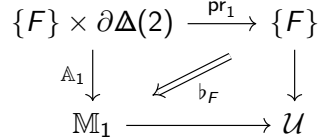
# The definition of curvature

## Definition (cont.)

An extension consists of $M_2$-many extensions to faces:



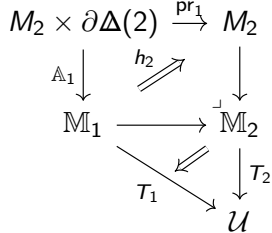

Here's the outer square for a single face $F$:



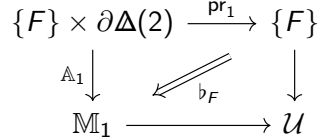
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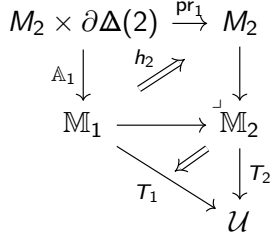



$T_1(\partial(F))$ is the curvature at the face $F$ and the filler $\flat_F : \mathsf{id} = T_1(\partial F)$ is called a flatness structure for the face $F$.

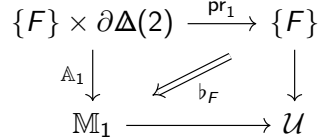
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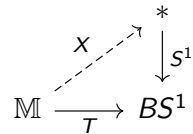
The distinction between the path $\flat_F$ and the endpoint $T_1(\partial(F))$ is small enough to be confusing.

# Vector fields

# Vector fields

Let $T : \mathbb{M} \to BS^1$ be an oriented tangent bundle on a 2-dim realization of a combinatorial manifold.

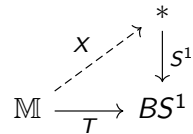
- Our bundles of mere circles can only model nonzero tangent vectors.
- A global section of this family would be a trivialization of $T$, so that's not a good definition.



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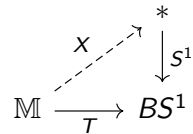



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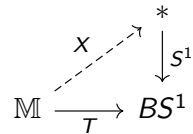

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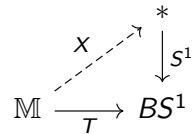
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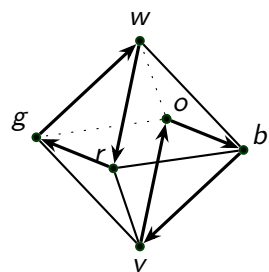
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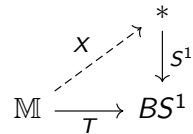


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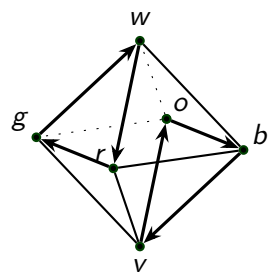
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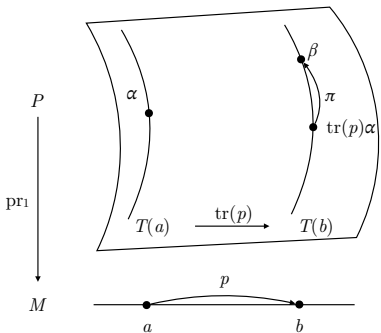



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- We model classical zeros by omitting the faces.

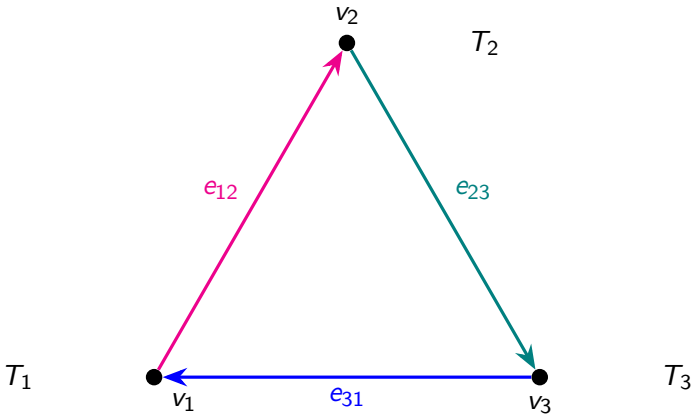


# Reminder: pathovers

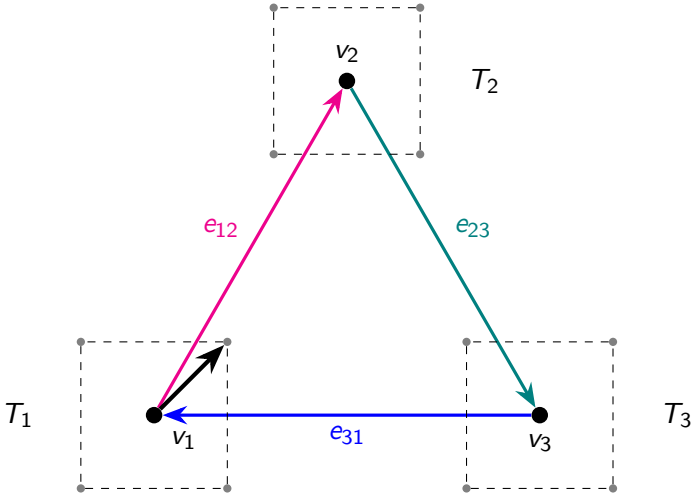



- Recall pathovers (dependent paths).
- There is an asymmetry: we pick a fiber to display $\pi$, the path over $p$.
- Dependent functions map paths to pathovers: $\mathsf{apd}(X)(p) : \mathsf{tr}_p(X(a)) = X(b)$ (simply denoted $X(p)$).

Next goal: define the index of a vector field on a face by computing $X(\partial F)$ around a face.

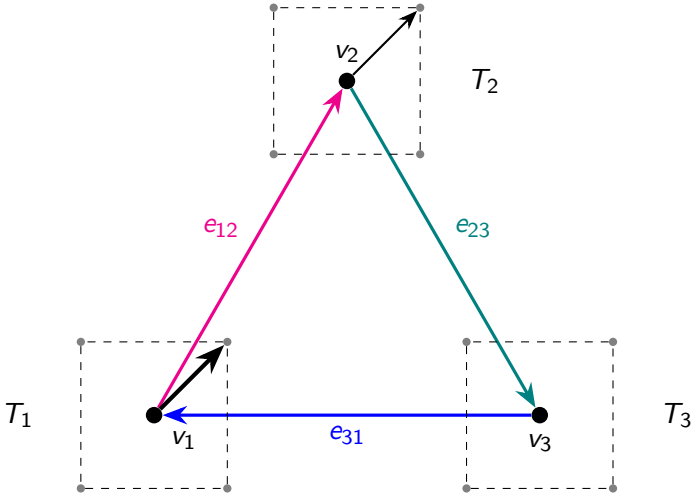



An example of swirling and index at this face.



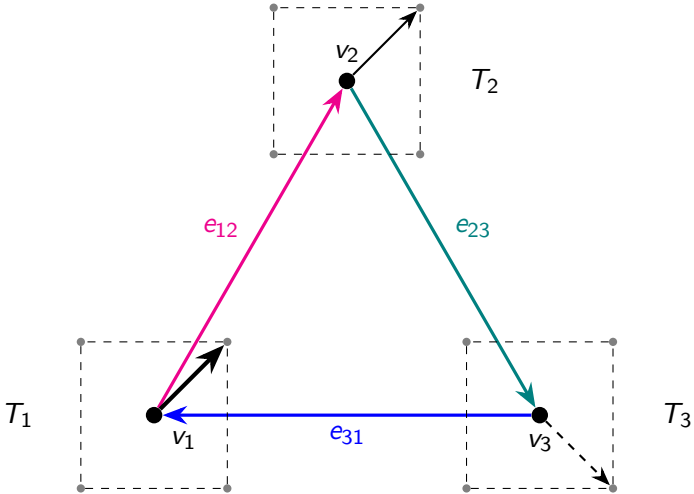

An example of swirling and index at this face.

- Denote by $X_1$ this vector $X(v_1) : T_1$.
-
-
-



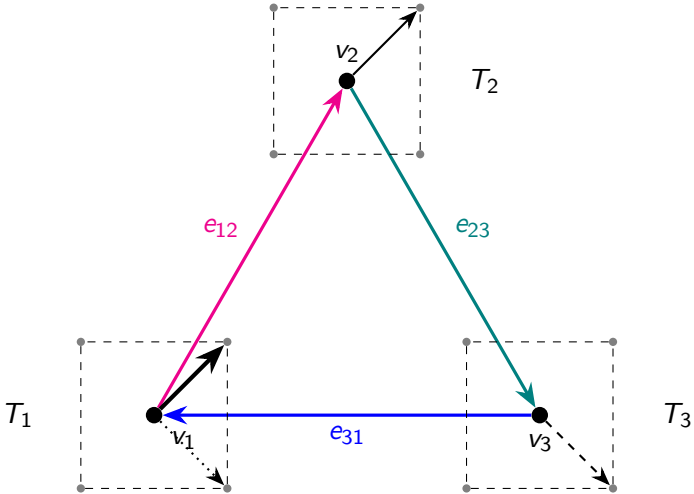

An example of swirling and index at this face.

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- Say $T_{21}$ is trivial. Denote the transported vector as thinner.
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-



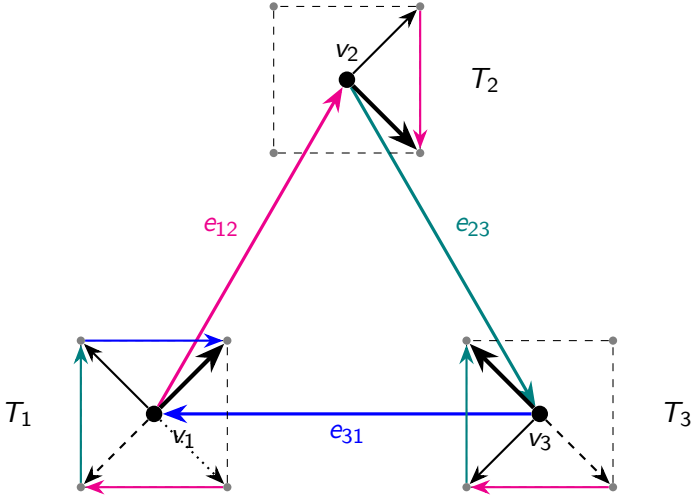

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- Denote by $X_1$ this vector $X(v_1) : T_1$.
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- Say $T_{32}$ rotates clockwise. Denote the twice-transported vector as dashed.
-

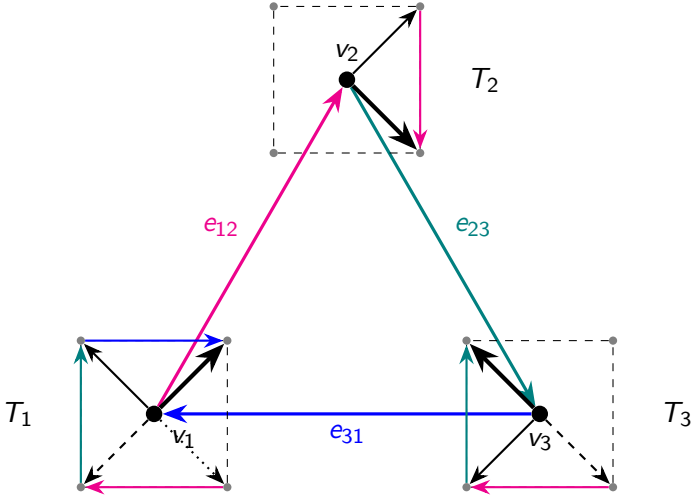



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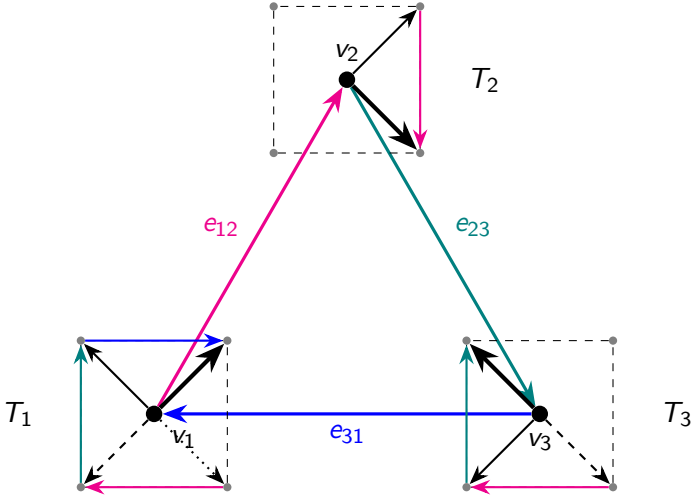
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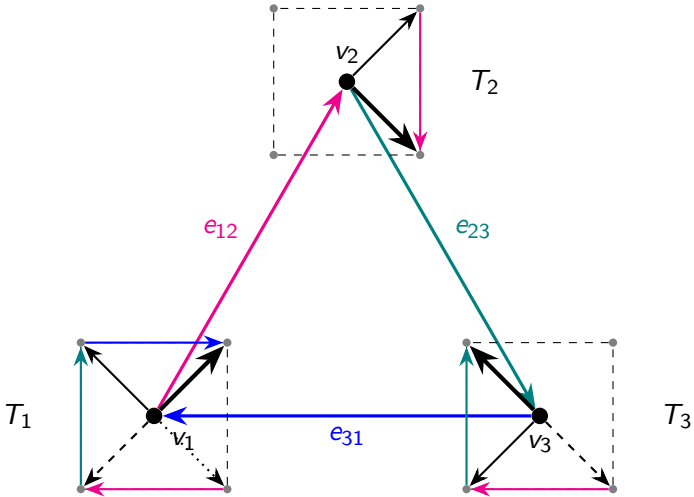

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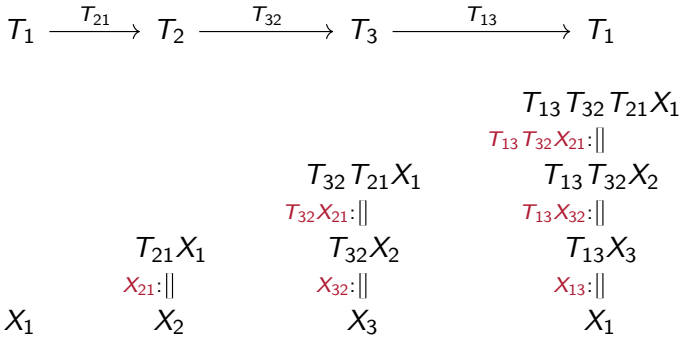



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- We translated all pathover data to the end of the loop.
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- The total pathover $X(\partial F)$ is called the swirling $X_F$ of $X$ at the face $F$.

# Symbolic version



# Index

$$\mathrm{tr}_F \stackrel{\mathrm{def}}{=} \mathrm{tr}(\partial F) \quad : T_1 =_{BS^1} T_1 \qquad \text{curvature}$$

$$\flat_F \stackrel{\mathrm{def}}{=} \flat(\partial F) \quad : \mathrm{id} =_{(T_1 =_{BS^1} T_1)} \mathrm{tr}_F \qquad \text{flatness}$$

$$X_F \stackrel{\mathrm{def}}{=} X(\partial F) \quad : \mathrm{tr}_F(X_1) =_{T_1} X_1 \qquad \text{swirling}$$

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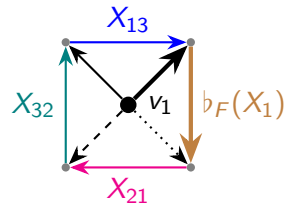
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## Definition

The flattened swirling of the vector field $X$ on the face $F$ is the loop

$$L_F^X \stackrel{\text{def}}{=} \flat_F(X_1) \cdot X_F : X_1 =_{T_1} X_1.$$

The index of the vector field $X$ on the face $F$ is the integer $I_F^X$ such that $\mathrm{loop}^{I_F^X} =_{S^1} \alpha(L_F^X)$ for some iso $\alpha : T_1 \simeq S^1$.



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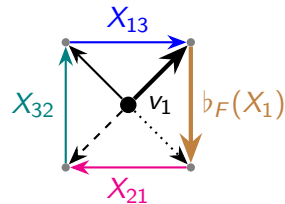
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The classical index uses a trivial local connection in a chart. We use the given type family and then subtract the effect of curvature.



# Main theorem

# Towards totaling

"Total swirling" is some complicated pathover. But the vector field removes some dependency.

# Pay off all our assumptions 1: torsor structure, vector field

$$T_1$$

$$T_{13}\,T_{32}\,T_{21}X_1$$

$$\downarrow T_{13}\,T_{32}X_{21}:$$

$$T_{13}\,T_{32}X_2$$

$$\downarrow T_{13}X_{32}:$$

$$T_{13}X_3$$

$$\downarrow X_{13}:$$

$$X_1$$

# Pay off all our assumptions 1: torsor structure, vector field

$T_1$

- Def: $\alpha_i \overset{\text{def}}{=} s(-, X_i) : T_i \xrightarrow{\sim} S^1$ (trivialization on 0-skeleton).

$T_{13}\, T_{32}\, T_{21} X_1$

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$X_{13}$: $\|$

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\begin{array}{c}
T_1 \\
\| \\
T_{13}\,T_{32}\,T_{21}X_1 \\
T_{13}\,T_{32}X_{21}:\ \| \\
T_{13}\,T_{32}X_2 \\
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X_1
\end{array}
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- Def: $\rho_{ji} \overset{\text{def}}{=} \alpha_j(T_{ji}(X_i))$ is the rotation of $T_{ji}$.

$$
\begin{array}{ccc}
T_i & \xrightarrow{T_{ji}} & T_j \\
{\scriptstyle \text{base} \mapsto X_i}\ \big(\ {\scriptstyle \alpha_i}\ \big\downarrow & & \big\downarrow\ {\scriptstyle \alpha_j}\ \big)\ {\scriptstyle \text{base} \mapsto X_j} \\
S^1 & \xrightarrow[(-)+\rho_{ji}]{} & S^1
\end{array}
$$

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T_1 \\
\\
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S^1 & \xrightarrow[(-)+\rho_{ji}]{} & S^1
\end{array}
$$

- Lemma: $\rho_{ij} = -\rho_{ji}$ because in $T_j$:
  $\rho_{ij} + \rho_{ji} + X_j = \rho_{ij} + T_{ji}X_i = T_{ji}(\rho_{ij} + X_i) = T_{ji}\,T_{ij}X_j = X_j.$

# Pay off all our assumptions 1: torsor structure, vector field (cont.)

$$T_1$$

$$T_{13}\,T_{32}\,T_{21}X_1$$

$$T_{13}\,T_{32}X_{21}:\ \|$$

$$T_{13}\,T_{32}X_2$$

$$T_{13}X_{32}:\ \|$$

$$T_{13}X_3$$

$$X_{13}:\ \|$$

$$X_1$$

# Pay off all our assumptions 1: torsor structure, vector field (cont.)

$T_1$

$T_{13}\,T_{32}\,T_{21}X_1$

$T_{13}\,T_{32}X_{21}$: ‖

$T_{13}\,T_{32}X_2$

$T_{13}X_{32}$: ‖

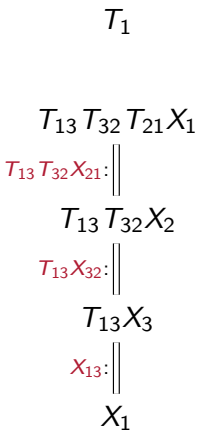
$T_{13}X_3$

$X_{13}$: ‖

$X_1$

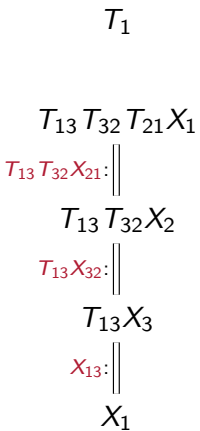
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  (compare $X_{ji} : T_{ji}X_i =_{T_j} X_j$)

# Pay off all our assumptions 1: torsor structure, vector field (cont.)



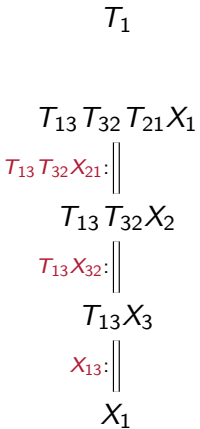

- Define $\sigma_{ji} \stackrel{\text{def}}{=} \alpha_j(X_{ji}) : \rho_{ji} =_{S^1} \text{base},.$ (compare $X_{ji} : T_{ji}X_i =_{T_j} X_j$)
- Paths of the form ($a =_{S^1} \text{base}$) can be added:
  - $+' : (a = \text{base}) \times (b = \mathit{base}) \to (a + b = \mathit{base})$.
  - $p +' q = (p + b) \cdot q$.

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$T_{13}\,T_{32}X_{21}$: $\|$

$T_{13}\,T_{32}X_2$

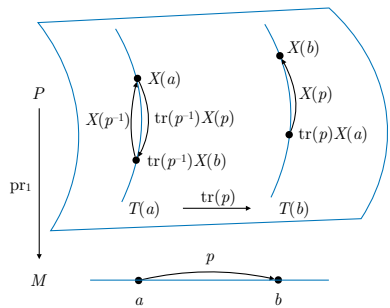
$T_{13}X_{32}$: $\|$

$T_{13}X_3$

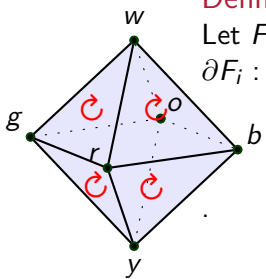
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# Pay off all our assumptions 2: no boundary, commutativity

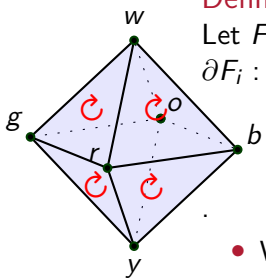



**Definition**

Let $F_1, \ldots, F_n$ be the faces of $\mathbb{M}$, $v_i : F_i$ be designated vertices, and $\partial F_i : v_i = v_i$ be the triangular boundaries. The total swirling is

$$X_{\text{tot}} \stackrel{\text{def}}{=} \sigma_{\partial F_1} +' \cdots +' \sigma_{\partial F_n}$$

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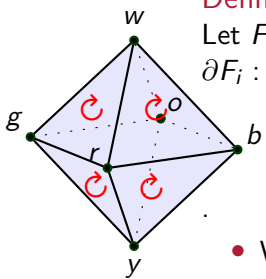

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- We assume that this expression involves every edge once in each direction.
- $S^1$ is commutative, hence complete cancellation.

# Consequence

$$\mathrm{tr}_F \stackrel{\mathrm{def}}{=} \mathrm{tr}(\partial F) \quad : T_1 =_{BS^1} T_1 \quad \text{curvature}$$

$$\flat_F \stackrel{\mathrm{def}}{=} \flat(\partial F) \quad : \mathrm{id} =_{(T_1 =_{BS^1} T_1)} \mathrm{tr}_F \quad \text{flatness}$$

$$X_F \stackrel{\mathrm{def}}{=} X(\partial F) \quad : \mathrm{tr}_F(X_1) =_{T_1} X_1 \quad \text{swirling}$$

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These can all be totaled in $S^1$ to give

$$\mathrm{tr}_{\mathrm{tot}} \overset{\text{def}}{=} \sum_i \rho_{\partial F} = \mathrm{base} \qquad X_{\mathrm{tot}} \overset{\text{def}}{=} \sum_i \sigma_{\partial F} = \mathrm{refl}_{\mathrm{base}}$$

$$\flat_{\mathrm{tot}} \overset{\text{def}}{=} \sum_i \flat_{\partial F} \qquad L_{\mathrm{tot}}^X \overset{\text{def}}{=} \sum_i \flat_{\partial F} + \sigma_{\partial F} = \sum_i \flat_{\partial F}$$

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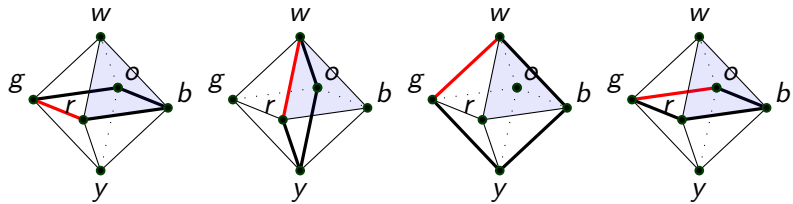
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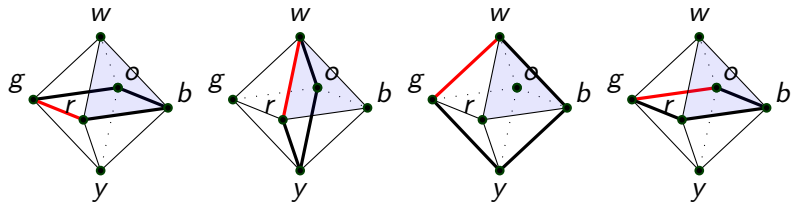
So in our lingo: the total flatness equals the total flattened swirling. □

# Examples

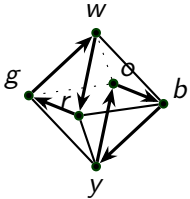



Each face contributes $\flat_F = H_R$, a 1/4-rotation. Total: 2.

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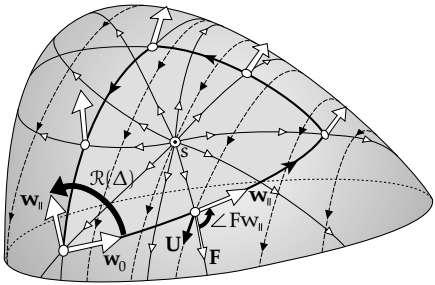



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For total index one obtains $+1$ from $F_{wrg}$, $+1$ from $F_{ybo}$, $+0$ from others. Total: 2.

# Classical proof




**[26.2]** *The difference* $\mathcal{R}(\Delta) - 2\pi\mathcal{I}_F(s)$ *can be found by summing over the edges* $K_j$ *the change* $\Phi(K_j)$ *in the illustrated angle* $\angle \mathrm{Fw}_{||}$*, i.e., the rotation of* $\mathbf{w}_{||}$ *relative to* $\mathbf{F}$.

Figure: Needham, T. (2021) Visual Differential Geometry and Forms.

- The classical proof is discrete-flavored.
- "$\angle Fw_{||}$" looked a lot like a pathover.
- Hopf's $\Phi$ is defined on edges, not loops. We imitated that too.

Thank you!

# Appendix: the octahedron is too special

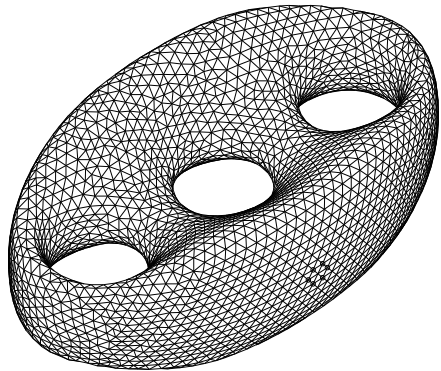
# Not everything is a regular solid

The octahedron is very symmetrical. The links are all squares and the tippings rotate by 90 degrees.

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A more typical situation has variable links, averaging to 6 sides. Also transport will rotate by small amounts.

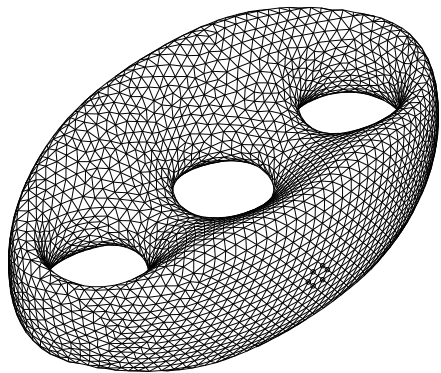


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The octahedron is very symmetrical. The links are all squares and the tippings rotate by 90 degrees.

A more typical situation has variable links, averaging to 6 sides. Also transport will rotate by small amounts.

Idea: introduce a subdivider $D_n$ which takes a $k$-gon to an $nk$-gon, adding $n$ vertices along each edge. Set $T_0(v) \stackrel{\text{def}}{=} D_{n(v)} \circ \text{link}(v)$. Make all tangent spaces the same LCM-sided polygon. Also transport should now be able to imitate small angles.



# Appendix: Conjectural dictionary

# Conjectionary

Homotopy realization likely amounts to the shape operator.

> *[G]iven a "cell complex" presentation of a classical topological space, if we can convert it into both a specification for a HIT and a colimit decomposition of that space that issufficiently "cofibrant", then ∫ will preserve that colimit and take the space to the HIT.*
>
> — Mike Shulman, *Brouwer's Fixed Point Theorem in Real-Cohesive HoTT*

# Conjectionary

Homotopy realization (and/or $\int$) may provide the following relationships.

| Classical octahedron | Homotopy realization, e.g. $\mathbb{O}$ |
|---|---|
| Classical combinatorial manifold $M$ | $\int M$ |
| Derivative | ap |
| Leibniz rule for $f, g : M \to \mathbb{R}$ | Given H-space $(A, *)$, $f, g : X \to A$, $p : x =_X y$ then $\mathrm{ap}(f * g)(p) = \mathrm{ap}(f)(p) * (ga) \cdot (fb) * \mathrm{ap}(g)(p)$. Because in $A \times A$, $(fp, gp) = (fp, \mathrm{refl}) \cdot (\mathrm{refl}, gp)$ |
| The sphere is not flat, as a pointwise statement | Nontrivial flatness on each face |

# Conjectionary

| | |
|---|---|
| Connections being "affine", and not (quite) 1-forms | $T_{ji} : T_i = T_j$ being a torsor and not (quite) a group |
| Space of connections for a given $P$ is contractible. | Two extensions to $\mathbb{O}_1$... |
| Tensor | ? |

# Conjectionary

| | |
|---|---|
| Maurer-Cartan form. | Hmm, consider the trivial connection on $M \times G$ or $G \to *$. |
| Gauge transformations acting on connections and maybe functions (YM) of connections. | |
| The based gauge group acts freely on connections. | |

# Conjectionary

| | |
|---|---|
| Characteristic classes. | $BS^1 \to B^n\mathbb{Z}$. |
| Chern-Weil theory. | $\mathbb{O} \xrightarrow{T} BS^1 \to B^n\mathbb{Z}$. |
| Hopf fibration. | $\mathbb{O} \xrightarrow{?} \mathrm{EM}(\mathbb{Z}, 1)$. |
| Zeros of $X$ = Poincare dual of the Euler class. | |